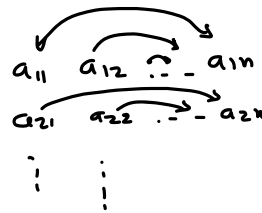


Homework 2

(a) left-right mirroring is equivalent to flipping each column of our RGB matrix

Then, we use `np.flip(image, axis=1)`
 ↑ This flips columnwise



(b) To partition $\{1, 2, \dots, 12\}$ into two groups

we do:

1. from the 12 numbers, in no particular order pick 6 numbers

$$\text{total ways to do this: } \binom{12}{6} = \frac{12!}{6!6!} = \frac{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 \cdot 7}{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2} = 924 \text{ ways}$$

G1:

2. We have to only build $(u_{j1}, u_{j2}, \dots, u_{j6})$

One group since the

$\Rightarrow$ G2:

other one follows

immediately from the previous one.

$(u_{j7}, \dots, u_{j12})$

Warning $\rightarrow$ We are counting twice since G_1, G_2 should be the same as G_2, G_1 . Then:

$$3. \text{ Divide by 2 our total number: } \frac{1}{2} \binom{12}{6} = \frac{924}{2} = 462 \text{ ways}$$

OK, how do we write each partition?

We are dividing our numbers in two groups: a binary

decision: (True or false). That is, dividing our group of $\{1, \dots, 12\}$

into two in the same as assigning a True/False value to each of them:

E.g.: G_{true}

1 2 3 4 5 6

G_{false}

7 8 9 10 11 12

corresponds

$[T, T, T, \dots, T, F, F, F, \dots, F]$

Write 1 if True, 0 if False: 1 1 1 1 1 0 0 0 0 0 0 0

$\Rightarrow$ This corresponds to a unique number in binary:
 $2^{11} + 2^{10} + 2^9 + 2^8 + \dots + 2^7 + 2^6$

We assign this partition this unique number.

Then, we can find all partitions according to all numbers that have a binary representation of 12 digits. (and contain 6 ones)

let's generate all this integers:

$$\text{integers} = \text{np.arange}(2^{12})$$

Next, we create a matrix s.t each row is the binary representation of this.

$$\text{binary} = \text{zeros}(2^{12}, 12)$$

Recall that to obtain the binary representation of a number, we continuously divide by 2 and record their remainder:

e.g.:

$$\begin{array}{r} 10 \overline{) 2} \\ \underline{0} 5 \overline{) 2} \\ \underline{1} 2 \overline{) 2} \\ \underline{0} 1 \overline{) 2} \\ \underline{1} 0 \end{array}$$

then save remainders in reverse order.

$$\begin{aligned} 10 &= 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0 \\ \Rightarrow 10 &= (1010)_{\text{binary}} \end{aligned}$$

Once we have obtained the binary representation of every number up to 2^{12} , we should realize that the only configurations we need are the ones who have exactly 6 1s (6 true values) since Gtrue needs exactly 6 numbers. Hence we choose the binary numbers s.t

The sum of their digits = 6.

Moreover, we impose the rule that the number 1 is

in `True` always.

(C) By layers, we mean the first dim in our $6 \times 6 \times 6$ array:
 $(\text{layer}, \text{row}, \text{column})$

So we have an array:

$[A[0], A[1], A[2], \dots, A[5]]$

then `np.min([...])` will take the minimum across all the array.

What if we specify an axis?

`np.min([...], axis=0)`

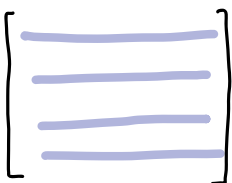
• `axis=0` is the layer axis, We compare (elementwise)

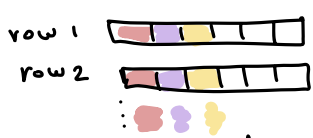
$A[0]$, with $A[1]$

So it returns a matrix of size (rows, columns).

(we read $\rightarrow$ )

• `axis=1` is the row axis. We compare:

$A[0] =$ 

Each element of each row:  results in a vector

is compared elementwise to find

and we do this for all $A[i]$. Hence we get a 6×6 matrix.

• `axis=2` is the column axis. The process is similar to before.

$\Rightarrow$ If we want minimum of each layer, we should check in both directions `axis=(1,2)`.

(d) Take a stochastic matrix $(20, 10)$.

Find the row who has minimal average distance.
w.r.t. other rows

Then compute:

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \vdots \\ v_{20} \end{bmatrix}$$

$$\text{row}_1 Av = \frac{1}{20} (\|v_1 - v_2\| + \|v_1 - v_3\| + \dots + \|v_1 - v_{20}\|)$$

$$\text{row}_2 Av = \frac{1}{20} (\|v_2 - v_1\| + \|v_2 - v_3\| + \dots)$$

$$\vdots$$

$$\text{row}_i Av = \frac{1}{20} \left(\sum_{j \neq i}^{20} \|v_i - v_j\| \right)$$

Notice:

		d_{12}	d_{13}	d_{14}	...
row ₁ Av computer	:	$\ v_1 - v_2\ $	$\ v_1 - v_3\ $	$\ v_1 - v_4\ $	...
row ₂ Av computer		$\ v_2 - v_1\ $	$\ v_2 - v_3\ $	$\ v_2 - v_4\ $	...
row ₃ Av computer		$\ v_3 - v_1\ $	$\ v_3 - v_2\ $	$\ v_3 - v_4\ $	...
					...

We are repeating computations.

due to the symmetry of norm.

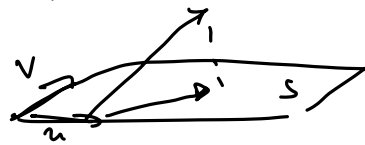
distance matrix $\rightarrow D =$

$$\begin{pmatrix} 0 & d_{12} & d_{13} & d_{14} & \dots & d_{120} \\ d_{12} & 0 & d_{23} & d_{24} & \dots & d_{220} \\ d_{13} & d_{23} & 0 & d_{34} & \dots & d_{320} \\ \vdots & & & \ddots & & \vdots \end{pmatrix} \rightarrow \text{np.sum}(T, \text{axis})$$

This is symmetric, diag 0!

$$\begin{pmatrix} d_{12} & d_{13} & d_{14} & \dots & d_{120} \\ d_{23} & d_{24} & \dots & & \\ d_{34} & \dots & & & \\ \vdots & & & & \end{pmatrix} \rightarrow \text{enough to compute this chunk.}$$

(e) If we project a vector:



How can we test the projection was done correctly?

We can check that the perpendicular part is orthogonal to each basis vector of the subspace we are projecting onto.